Please stick the barcode label here.

Answers written in the margins will not be marked.

*3.

Figure 3.1

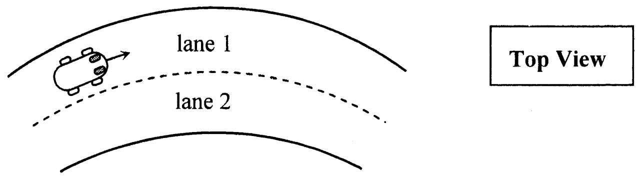


Figure 3.1 shows the top view of a horizontal road with two circular lanes. A car of mass 1200 kg moves with constant speed in lane 1 of radius 45 m.

(a) (i) Name the force that provides the centripetal force for the car. If the maximum value of this force is 8000 N, calculate the highest speed of the car such that it can keep in lane 1. (3 marks)

....

....

....

....

....

....

....

(ii) Suppose the car takes lane 2 instead of lane 1 and the maximum value of the force providing the centripetal force is still 8000 N. Would the car's highest speed in lane 2 be smaller than, larger than or the same as that found in (a)(i) ? Explain. (2 marks)

....

....

....

....

....

Answers written in the margins will not be marked.

4. Train A initially travels at a speed of 60 m s^{-1} along a straight horizontal railway. Another identical train B travels ahead of A in the same direction on the same railway. Due to mechanical failure, B is only travelling at 20 m s^{-1} (Figure 4.1).

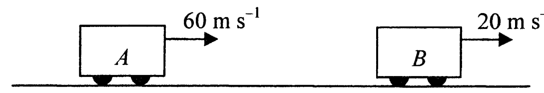


Figure 4.1

At time $t = 0$, A and B are x m apart, the captain of A receives a stopping signal and immediately A decelerates at 4 m s^{-2} while B continues to travel at 20 m s^{-1}. A eventually collides with B after 5 s. Neglect air resistance.

(a) (i) Find the speed of A just before collision. (2 marks)

....

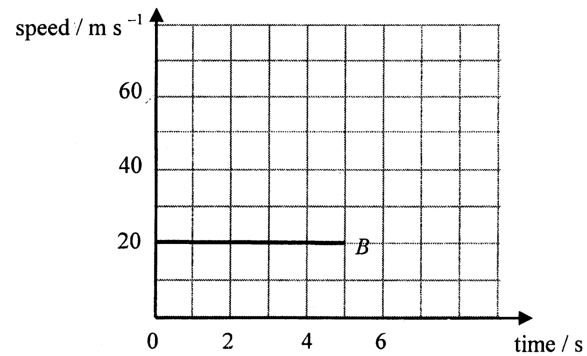
....

....

....

....

(ii) The graph below shows how the speed of B varies with time within this 5 s. Sketch on the same graph the variation of the speed of A within the same period. (1 mark)



(iii) Based on the above information, determine the separation x of the two trains at $t = 0$. (3 marks)

....

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Please stick the barcode label here.

(a) As shown in Figure 1.1b, the bulb of the soil thermometer is very large compared to those of common thermometers. Suggest a reason for this design. (1 mark)

(b) On a certain morning, the air temperature is 15°C. An observer takes a measurement of the soil temperature at 1 m deep. The thermometer reading is 20°C. It is given that the mass of the paraffin wax enclosing the thermometer bulb is 0.015 kg, and the specific heat capacity of paraffin wax is 2.9×10^3 J kg^{-1} °C^{-1}.

(i) Calculate the energy loss of the paraffin wax as it cools down to the air temperature. (2 marks)

(ii) It is known that the paraffin wax enclosing the bulb of the thermometer gains or loses energy at a constant rate of 0.5 J s^{-1}, estimate the time taken for the paraffin wax to reach the air temperature after the thermometer is lifted out of the soil. (2 marks)

(iii) If there is no paraffin wax enclosing the bulb of the thermometer, explain how the thermometer reading as recorded by the observer is affected. (2 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

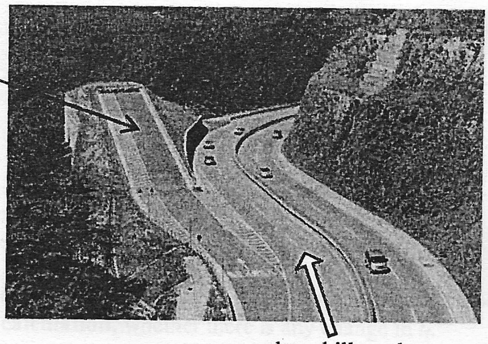
Provided by dse.life

Please stick the barcode label here.

(b) (i) It is known that the brake pads would provide much smaller frictional forces or even brake failure would occur (i.e. without any friction provided) when reaching a temperature of over a few hundred degrees Celsius. Explain why it is not recommended for a vehicle driver to **apply the brakes continuously** during a long downhill trip. (1 mark)

(ii) The figure shows an emergency escape ramp (slanting upwards) built for stopping vehicles with brake failure resulting from the situation described in (b)(i). If such a ramp makes an angle of 30° with the horizontal and a vehicle with brake failure enters the bottom end of the ramp at a speed of 25 m s^{-1}, estimate how far it will travel along the ramp before it stops. Neglect air resistance and mechanical resistances within the vehicle. ($g = 9.81$ m s^{-2}) (2 marks)

emergency escape ramp



downhill road

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Provided by dse.life

4. Figures 4.1 (a) and (b) show the cross-section of a fixed spring gun fitted with a small cannon ball.

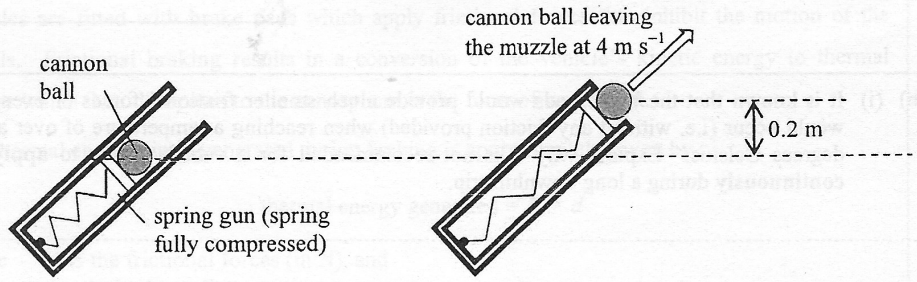


Figure 4.1(a) Figure 4.1(b)

The fully compressed light spring (Figure 4.1(a)) is released so that the cannon ball of mass 0.3 kg leaves the muzzle of the gun at a speed of 4 m s^{-1} (Figure 4.1(b)). Neglect air resistance. (g = 9.81 m s^{-2})

(a) During the process from the time when the spring is fully compressed till the cannon ball just leaves the muzzle,

(i) how much energy is transferred from the spring to the cannon ball ? (3 marks)

....

....

....

....

....

....

....

....

(ii) explain whether the total momentum of the spring gun and the cannon ball is conserved. (2 marks)

....

....

....

....

....

....

Answers written in the margins will not be marked.

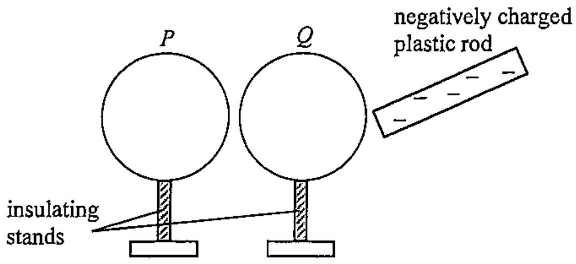
Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Provided by dse.life

9. Figure 9.1 shows two identical conducting light spheres P and Q placed close to each other without touching.

Figure 9.1



Both spheres are uncharged initially. A negatively charged plastic rod is brought close to Q but without touching the sphere.

(a) On Figure 9.1, sketch the distribution of charges on P and on Q. (2 marks)

A student touches sphere P momentarily with his finger while the plastic rod is in the position as shown. The plastic rod is removed afterwards.

(b) Explain whether the amount of charges on the plastic rod would decrease. (2 marks)

....

....

....

....

....

(c) **The two light spheres** are now suspended by nylon threads from fixed supports such that the spheres are far apart from each other. Explain how you would use one of the spheres to test for the unknown polarity of another charged rod. (2 marks)

....

....

....

....

....

....

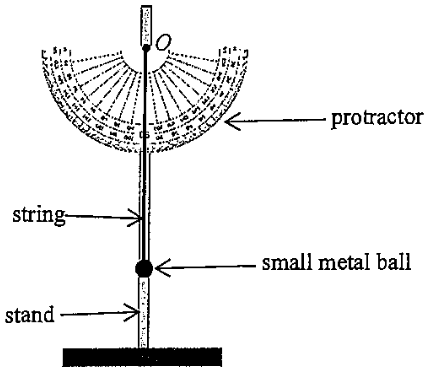
Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

10. (a) As shown in Figure 10.1, one end of an inextensible string is attached to a small metal ball while its upper end O is fixed to a stand. A protractor is fixed onto the stand behind the string.

Figure 10.1



Write down the steps that you would take to demonstrate the conservation of energy using the set-up given. (4 marks)

....

....

....

....

....

....

....

....

....

....

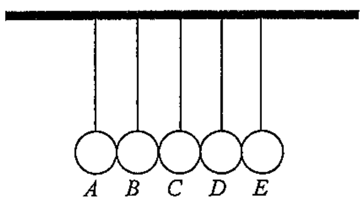
....

....

....

(b) Figure 10.2 shows a Newton's cradle which consists of five identical metal spheres suspended in a row by inextensible light strings of equal length from a fixed horizontal support.

Figure 10.2



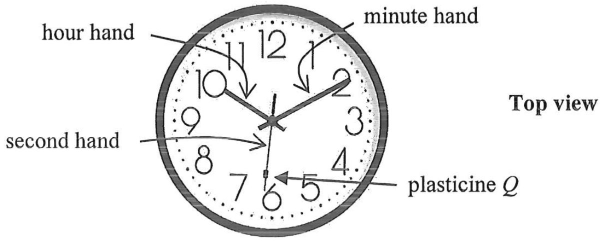
Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

*4. A large clock facing upwards is placed horizontally on a table as shown. A small piece of plasticine Q rests on the steadily rotating second hand of the clock.

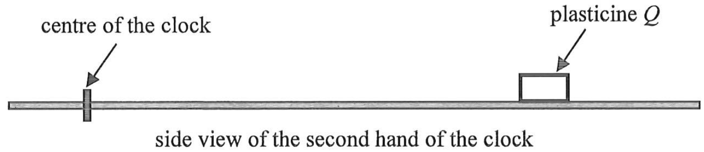
Figure 4.1



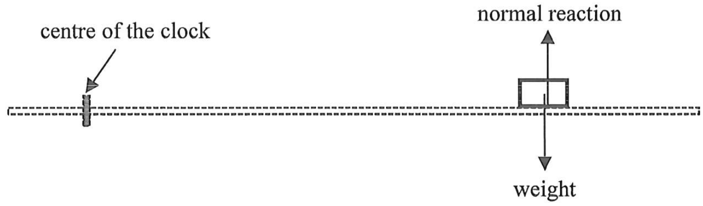
(a) Find the angular speed ω, in rad s^{-1}, of the second hand of the clock. (1 mark)

(b) Figure 4.2 shows the side view of the second hand of the clock.

Figure 4.2



In the free-body diagram of plasticine Q below, add a force acting on Q to complete the diagram and name this force. (1 mark)



(c) A student claims that the centripetal force required for Q will become smaller if it is placed nearer to the centre of the clock. Explain briefly whether this claim is correct or not. (2 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

(b) (i) Find the average kinetic energy of the gas molecules. (2 marks)

(ii) The mass of one mole of neon gas is 0.020 kg. Find the root-mean-square speed of the gas molecules. (2 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

(a) Find the tension of the string. (2 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.